

Universidad Autónoma de Nuevo León
Facultad de Ingeniería Mecánica y Eléctrica

Maestría en Ciencias de la Ingeniería Eléctrica
Unidad de Aprendizaje: **Control No Lineal**

Actividad Fundamental 4

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Simulation environment

We implemented a simulation of the model recovered from the state-space characterization article using the MuJoCo physics engine for Python. The simulation serves as a virtual test bed where the proposed trajectories can be validated under the dynamic model. The MuJoCo Python bindings allow us to define the context of our simulation from an XML string by calling the `mujoco.MjModel.from_xml_string(MODEL_XML)` method. We extend from MuJoCo's `<worldbody>` tag on which we place a root `<body>` tag, representing our base frame; this allows us to organize the machine in the simulation using the three-chain hierarchical tree.

The simulation environment is defined through several specialized XML attributes that handle different aspects of the physical world. Table 1 summarizes the function of each tag within the context of the lithography scanner model, and Table 2 maps the specific names used in the MuJoCo XML to their counterparts in the structural diagram of the multi-body lithography model.

Table 1: MuJoCo XML tag functions in the scanner simulation.

Tag	Function
<code><worldbody></code>	Defines the inertial root, the floor plane ($\mathcal{F}_0$).
<code><body></code>	Creates a rigid body with its own local coordinate system.
<code><inertial></code>	Assigns mass m_{i_j} and the aggregated inertia tensor I_{i_j} .
<code><joint></code>	Implements the degrees of freedom f_{i_j} and $\theta_{z_{i_j}}$.
<code><geom></code>	Defines the material (granite, silicon, glass) and collision shapes.
<code><actuator></code>	Models the Lorentz and moving-coil motors via servo controllers.

Table 2: Mapping of MuJoCo XML bodies to the model components.

<code><body></code> name	Model body
<code>base_frame</code>	Base Frame ($\mathcal{F}_1$)
<code>w_ls_act</code>	Wafer LS ($i_3 = 2$)
<code>w_ls_flex</code>	flex coupling ($i_3 = 3$)
<code>w_ss_act</code>	Wafer SS ($i_3 = 4$)
<code>w_ss_flex</code>	flex coupling ($i_3 = 5$)
<code>w_st_act</code>	Wafer Stage ($i_3 = 6$)
<code>wafer</code>	Die ($N_3 = 7$)
<code>r_ls_act</code>	Reticle LS ($i_1 = 2$)
<code>r_ls_flex</code>	flex coupling ($i_1 = 3$)
<code>r_ss_act</code>	Reticle SS ($i_1 = 4$)
<code>r_ss_flex</code>	flex coupling ($i_1 = 5$)
<code>r_st_act</code>	Reticle Stage ($i_1 = 6$)
<code>mask</code>	Mask ($N_1 = 7$)
<code>metro_frame</code>	Metrology Frame ($i_2 = 2$)
<code>optics_box</code>	Optics Box ($i_2 = 3$)
<code>lens</code>	Lens Element ($i_2 = 4$)

Topology and Kelvin-Voigt couplings

The simulation topology mirrors the multi-body hierarchy described in the modeling chapter. The common base frame ($\mathcal{F}_1$) is anchored to the world frame ($\mathcal{F}_0$) via a 3-DOF joint (two prismatic, one revolute) representing the isolation system with stiffness K_1 and damping C_1 . From this base, three serial chains are constructed using nested body definitions in MuJoCo’s XML format, where each child body’s joint coordinate corresponds directly to the relative displacement f_{i_j} and rotation $\theta_{z_{i_j}}$ defined in the kinematics model.

The mechanical properties of the couplings, represented by the Kelvin-Voigt elements, are implemented via the **stiffness** and **damping** attributes of the `<joint>` tags. For actuated stages, where the physical stiffness is replaced by the controller’s active response, the simulation uses high **kp** values. For the actuated stages (Long-Stroke, Short-Stroke, and Fine stages), we use MuJoCo’s position actuators. To approximate the near-infinite stiffness of a rigid servo motor while maintaining numerical stability with the implicit integrator at a timestep of $dt = 10^{-4}$ s, we employ a proportional gain $K_P = 10^7$, with derivative gains K_V tuned to the C/K ratios provided in the characterization data.

Chain parameters

Tables 3, 4, and 5 illustrate the distribution of the stiffness and damping variables for the three chains.

Table 3: Wafer chain stiffness (K) and damping (C) parameters.

Coupling Interface	Stiffness (K)	Damping (C)
Floor $\rightarrow$ Base Frame	K1	C1
Base $\rightarrow$ LS Stage	KP_ACT	KV_ACT_TRANS
LS Stage $\rightarrow$ LS Flex	K_LS_F	C_LS_F
SS Stage $\rightarrow$ SS Flex	K_SS_F	C_SS_F
Fine Stage $\rightarrow$ Flex	K_ST_F	C_ST_F
Stage $\rightarrow$ Die	K8_W	C8_W

Table 4: Reticle chain stiffness (K) and damping (C) parameters.

Coupling Interface	Stiffness (K)	Damping (C)
Floor $\rightarrow$ Base Frame	K1	C1
Base $\rightarrow$ LS Stage	KP_ACT	KV_ACT_TRANS
LS Stage $\rightarrow$ LS Flex	K_LS_F	C_LS_F
SS Stage $\rightarrow$ SS Flex	K_SS_F	C_SS_F
Fine Stage $\rightarrow$ Flex	K_ST_F	C_ST_F
Stage $\rightarrow$ Mask	K8_R	C8_R

Table 5: Optics chain stiffness (K) and damping (C) parameters.

Coupling Interface	Stiffness (K)	Damping (C)
Base Frame $\rightarrow$ Metro Frame	K2_0	C2_0
Metro Frame $\rightarrow$ Optics Box	K3_0	C3_0
Optics Box $\rightarrow$ Lens Element	K4_0	C4_0

Step-and-scan controller

The simulation incorporates a **StepperController** that generates the references for the scanning and stepping phases. During the scanning phase, the wafer stage moves at a constant velocity $v_{w,\text{scan}} = 0.8 \text{ m/s}$ while the reticle stage moves in the opposite direction at $v_{r,\text{scan}} = 3.2 \text{ m/s}$, maintaining the demagnification ratio $\alpha = 0.25$. This testing framework allows us to evaluate the impact of the centripetal and Coriolis terms and the effectiveness of the inertia aggregation when subjected to high-acceleration ($a_{w,\text{max}} = 20 \text{ m/s}^2$) transients.